Appendix C

Kirkwood-Buff relations

C.1 Structure factor for mixtures

In section 5.1.7.1 we discussed the relation between the structure factor $S(q)$, Eq. (5.1.40), of a one-component system and the mean-squared value of the Fourier transform of the particle density (Eq. (5.1.38)):

$$\rho(\mathbf{q}) = \sum_{i=1}^{N} \mathrm{e}^{i\mathbf{q}\cdot\mathbf{r}_i} = \int_V \mathrm{d}\mathbf{r}\, \rho(\mathbf{r}) \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}} \, .$$

For an n-component system, we can generalize this relation to yield expressions for the partial structure factors $S_{ab}(\mathbf{q})$ that measure the cross-correlations between fluctuations in the Fourier transforms of the densities of species a and b:

$$\begin{aligned} S_{ab}(\mathbf{q}) &= \frac{1}{\sqrt{\langle N_a \rangle \langle N_b \rangle}} \langle \delta\rho_a(\mathbf{q}) \delta\rho_b(-\mathbf{q}) \rangle \\ &= \frac{1}{\sqrt{\langle N_a \rangle \langle N_b \rangle}} \int_V \int_V \mathrm{d}\mathbf{r}\, \mathrm{d}\mathbf{r}' \left\langle \delta\rho_a(\mathbf{r}) \delta\rho_b(\mathbf{r}') \right\rangle \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')} \, , \end{aligned} \tag{C.1.1}$$

where $\delta\rho(\mathbf{q})_i$ denotes the fluctuation of $\rho(\mathbf{q})$ around its average value. If we take the limit $q \to 0$ in the second half of Eq. (C.1.1), we get:

$$\lim_{q\to 0} S_{ab}(\mathbf{q}) = \frac{\langle \Delta N_a \Delta N_b \rangle}{\sqrt{\langle N_a \rangle \langle N_b \rangle}} \tag{C.1.2}$$

where ΔN_a denotes the fluctuation in the total number of particles of species a in the system. Following Kirkwood and Buff [717], we show that Eq. (C.1.2) has a direct thermodynamic interpretation, which provides a powerful route to determine the composition-dependence of chemical potentials in solution. For a multi-component system, we can generalize the expression for the Grand-Canonical partition, Ξ, Eq. (2.3.19) to

$$\Xi(\{\mu\}, V, T) \equiv \sum_{N_1, N_2, \cdots, N_n = 0}^{\infty} \prod_{a=1}^{n} \exp(\beta\mu_a N_a) \, \mathrm{e}^{-\beta F(\{N\}, V, T)}, \tag{C.1.3}$$

where $\{\mu\}$ denotes $\mu_1, \mu_2, \cdots, \mu_n$ and $\{N\}$ stands for $N_1, N_2, \cdots, N_n$. The Grand Potential, $\Phi = \Phi(\{\mu\}, V, T)$ is given by

$$\Phi = -k_B T \ln \Xi(\{\mu\}, V, T) \, .$$

From Eq. (C.1.3), it then follows that

$$\left(\frac{\partial \Phi}{\partial \mu_a}\right) = -\langle N_a \rangle \tag{C.1.4}$$

and

$$\left(\frac{\partial^2 \Phi}{\partial \mu_a \partial \mu_b}\right) = -\left(\frac{\partial N_a}{\partial \mu_b}\right) = -\beta \langle \Delta N_a \Delta N_b \rangle_{\{\mu\},V,T} . \tag{C.1.5}$$

Comparing Eq. (C.1.5) with Eq. (C.1.2) indicates that there is a close relation between the behavior of the structure factors $S_{ab}(q)$ in the limit $q \to 0$, and the thermodynamic derivatives of the Grand Potential.

Kirkwood and Buff were the first to propose these relations [717]. However, they also expressed their results in terms of integrals over the radial distribution functions $g_{ab}(r)$, and many simulation studies use the $g(r)$-approach to compute the composition-dependence of chemical potential. The $g(r)$-based relations are correct in principle, but as explained below Eq. (5.1.42), they are very dangerous to use in simulations on small (and not even very small) systems. It is, therefore, better to stick with Eq. (C.1.2) [718].[1]

We still have to explain why the relations Eq. (C.1.5) are important. The key reason is that they allow us to compute the variation with the composition of the chemical potentials of the various species in a multi-component mixture under conditions where particle-insertion methods (see section 8.5.1) fail.

We first note that at constant T and V:

$$\mathrm{d}\Phi = -\sum_{r=1}^{n} N_r \mathrm{d}\mu_r$$

and, hence

$$\left(\frac{\partial \Phi}{\partial N_a}\right)_{T,V,N'} = -\sum_{r=1}^{n} \langle N_r \rangle \left(\frac{\partial \mu_r}{\partial N_a}\right)_{T,V,N'} ,$$

where N' and μ' denote the set of all $\{N\}$ and $\{\mu\}$ that are not being varied in that particular differentiation (i.e., we use the same symbol for different sets). It then follows that

$$\begin{aligned}\left(\frac{\partial^2 \Phi}{\partial N_a \partial N_b}\right)_{T,V,N'} = &\sum_{r,s=1}^{n} \left(\frac{\partial^2 \Phi}{\partial \mu_r \partial \mu_s}\right)_{T,V,\mu'} \left(\frac{\partial \mu_r}{\partial N_a}\right)_{T,V,N'} \left(\frac{\partial \mu_s}{\partial N_b}\right)_{T,V,N'} \\ &- \sum_{r=1}^{n} \langle N_r \rangle \left(\frac{\partial^2 \mu_r}{\partial N_a \partial N_b}\right)_{T,V,N'} .\end{aligned} \tag{C.1.6}$$

[1] Ref. [719] describes an alternative approach to exploit the Kirkwood-Buff relations in systems where the total numbers of molecules of each species are fixed.

From the Gibbs-Duhem equation ($d\Phi_{V,T} = -\sum_r N_r d\mu_r$) it follows that

$$\left(\frac{\partial^2 \Phi}{\partial N_a \partial N_b}\right)_{T,V,N'} = -\left(\frac{\partial \mu_a}{\partial N_b}\right)_{T,V,N'} - \sum_{r=1}^{n} N_r \left(\frac{\partial^2 \mu_r}{\partial N_a \partial N_b}\right)_{T,V,N'} . \quad \text{(C.1.7)}$$

Combining Eqs. (C.1.6) and (C.1.7), we get

$$\left(\frac{\partial \mu_a}{\partial N_b}\right)_{T,V,N'} = -\sum_{r,s=1}^{n} \left(\frac{\partial^2 \Phi}{\partial \mu_r \partial \mu_s}\right)_{T,V,\mu'} \left(\frac{\partial \mu_r}{\partial N_a}\right)_{T,V,N'} \left(\frac{\partial \mu_s}{\partial N_b}\right)_{T,V,N'} . \quad \text{(C.1.8)}$$

Note that Eq. (C.1.8) has the form of a matrix equation

$$A_{ab} = A_{ar} B_{rs} A_{sb} ,$$

where

$$A_{ab} = A_{ba} = \left(\frac{\partial \mu_a}{\partial N_b}\right)_{T,V,N'} ,$$

and

$$B_{rs} = B_{sr} = \beta \left\langle \Delta N_r \Delta N_s \right\rangle_{\{\mu\},V,T} .$$

But then, $\mathbf{A} = \mathbf{B}^{-1}$ or, in compact notations,

$$\beta \left(\frac{\partial \mu_r}{\partial N_b}\right)_{T,V,N'} \langle \Delta N_b \Delta N_s \rangle_{T,V,\mu} = \delta_{rs}, \quad \text{(C.1.9)}$$

where δ_{rs} is the Kronecker δ-function. In other words, once we have computed the matrix of cross-correlations in the number fluctuations, then we know the variation with the composition of the chemical potentials of all species. Note that the A-matrices refer to an ensemble where T, V, and N' are fixed, whereas B expresses the fluctuations in an ensemble at constant μ, V, T.

C.2 Kirkwood-Buff in simulations

To compute the phase behavior of mixtures, we need to know the variation of the chemical potentials with composition. For dense liquids, we cannot compute μ_i using the particle-insertion method (8.5.1), or using grand-canonical simulations (6.5) because the probability of successful insertions becomes very small. It is under such conditions that Eq. (C.1.2) becomes useful: if the number of particles of the different species is fixed, $S_{ab}(\mathbf{q} = 0)$ vanishes identically, but we can compute the limit $S_{ab}(\mathbf{q} = 0)$ for $q \to 0$, and this limit is well defined and, apart from possible finite size effects, equal to the desired value.